05 · What to control for — and what NOT to (pathmc)

July 13, 2026

The business decision behind the statistics. Every effect estimate in every other notebook depends on a choice made *before* any model runs: **which variables to control for** (put in the model / adjust for). Get it wrong and the number is biased no matter how fancy the estimator. The instinct "throw in every column we have" is actively dangerous. This is the notebook that lets you tell a skeptical CMO *exactly why* you controlled for what you did.

The four roles a variable can play — and the rule for each

To "**control for**" (equivalently *adjust for*, *condition on*) a variable W means to hold it fixed when comparing treated vs untreated — e.g. by including it in the regression. Whether that helps or hurts depends entirely on W's causal role between treatment T and outcome Y:

- **Confounder** — a common cause of *both* T and Y (e.g. customer loyalty drives both who gets emailed and how much they spend). It opens a spurious "backdoor" path $T \leftarrow W \rightarrow Y$. $\rightarrow$ **Control for it.** Failing to is the classic omitted-variable bias.
- **Mediator** — a variable *on the causal path* $T \rightarrow W \rightarrow Y$ (the effect flows through it). $\rightarrow$ **Do NOT control for it** — you'd subtract out part of the very effect you're trying to measure.
- **Collider** — a common *effect* of T and Y ($T \rightarrow W \leftarrow Y$). Conditioning on a collider *opens* a path that was closed, **manufacturing** a correlation that isn't causal. $\rightarrow$ **Do NOT.**
- **M-bias collider** — a *pre-treatment* variable that is a collider between two hidden causes. $\rightarrow$ **Do NOT** — proof that "just control for everything measured before treatment" is wrong.

The tool that decides this formally is the **backdoor criterion** on the **DAG** (the causal graph): a valid **adjustment set** is any set of variables that blocks every backdoor path from T to Y *without* including a mediator or a descendant of T. `pathmc` reads the graph and enumerates the valid sets for us.

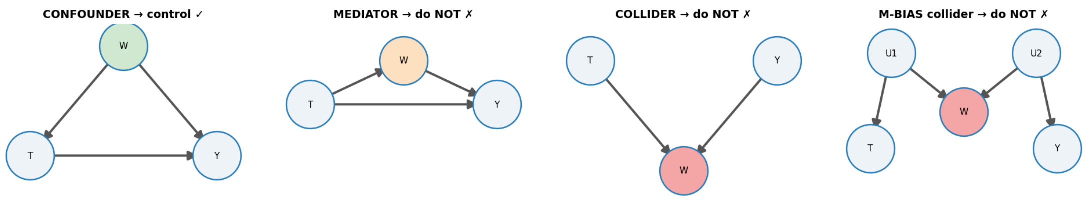
What this notebook does

(a) a **gallery** of the four structures with the correct action — and, right after it, a working simulation for the two that usually stay pictures (mediator over-control and M-bias); (b) we formalize d-separation and the backdoor criterion, apply them **by hand**, and only then let `pathmc` enumerate the admissible adjustment set and flag colliders straight from the DAG; (c) we show the bias of each wrong choice **in euros** — predicted from the graph first, confirmed in simulation second, including the disguised version everybody commits (filtering to responders); (d) we bound the residual risk from an *unobserved* confounder with a sensitivity contour and an **E-value** (the minimum confounder strength that could explain the effect away), read the full **posterior** of the effect, and close with real data (LaLonde), a practitioner's covariate checklist and a one-paragraph decision memo.

It follows the cookbook's standard **7-step arc**, with **identification** (which variables make the effect recoverable) — not estimation — as the whole point. That claim is put on trial twice: **Step 0** runs the same plain OLS under the naive and the backdoor-adjusted control sets (the estimator is held fixed; only the DAG's advice changes, and the answer moves by €3), and **5x** runs the same Bayesian posterior under the right and the wrong adjustment sets (the identification is what changes, and the *decision* flips). The estimator is never the thing that saves you.

1b · The four structures — a control-or-not gallery

Every "should I control for W?" question reduces to *what role W plays* between treatment T and outcome Y:



- **Confounder** (common cause of T and Y): **control** — it opens a backdoor path.
- **Mediator** (on the causal path T→W→Y): **do not** — you'd remove part of the effect you want.
- **Collider** (common effect of T and Y): **do not** — conditioning *opens* a spurious path.
- **M-bias collider** (pre-treatment, common effect of two latents that separately cause T and Y): **do not** — even a *pre-treatment* variable can be a trap. "Adjust for everything pre-treatment" is wrong.

1c · The other two traps, with data — the mediator first

The gallery's confounder and collider get a full simulation in §2–5. The other two panels are the ones that usually stay pictures — so before moving on, we simulate **both** in miniature, and every rule in the gallery has a number attached.

Trap 1 — the mediator. A discount email lifts spend two ways: directly (the customer redeems from the email itself) and *indirectly*, by pulling the customer to the site where they spend more. Site visits sit *on the causal path* — a **mediator**. We deliberately **randomize** T (a coin-flip email) so there is no confounding anywhere: whatever goes wrong below is purely the control choice. The data-generating model, with $\sigma(z) = 1/(1+e^{-z})$:

$$
\begin{aligned}
T &= \text{email} \sim \text{Bernoulli}(0.5) && \text{(randomized — no confounding by design)} \\
M &= \text{visited} \sim \text{Bernoulli}(\sigma(-0.5 + 1.5\,T)) && \text{(the email drives site visits)} \\
Y &= \text{spend} = 20 + 3\,T + 8\,M + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,\,4^2) && \text{(spend responds to both)}
\end{aligned}
$$

The number the CMO is asking for — "what does one email add to spend, *all channels included*?" — is the **total effect**, which splits into a direct and an indirect (through-visits) piece:

$$\tau_{\text{total}} = \underbrace{3}_{\text{direct}} + \underbrace{8\,[P(M{=}1 \mid T{=}1) - P(M{=}1 \mid T{=}0)]}_{\text{indirect, via visits}} = 3 + 8\,[\sigma(1.0) - \sigma(-0.5)] \approx 5.8.$$

Two regressions on the same simulated data — one with no controls, one “helpfully” controlling for `visited`:

```
TRUTHS   direct = €3.00  *  total = 3 + 8*DeltaM = 3 + 8*0.348 ~ €5.78
no controls           : €5.69   -> the TOTAL effect (what the CMO asked for)
controlling 'visited' : €2.92   -> only the DIRECT effect - the €2.78 that flows
through visits is silently subtracted
```

Read-out (mediator). With T randomized, the no-controls regression is unbiased for the total effect — and lands on it (up to sampling noise around the ≈ €5.8 truth). Adding `visited` does not “improve precision”; it **changes the estimand** (the quantity you are actually estimating — here shifting the target from the total effect to the direct effect): the coefficient now answers *“what would the email do if we could freeze site visits?”* — the direct effect (≈ €3), not the number the CMO asked about. Sometimes that split is exactly what you want — notebook 04 builds the total/direct/indirect machinery deliberately. The sin is not computing a direct effect; it is computing one *by accident* and calling it the campaign’s lift: an analyst who controls for post-send engagement quietly books only the fraction of the lift that bypasses the site.

Trap 2 — M-bias: the pre-treatment variable that is still a collider. Two customer traits we never observe: *deal-proneness* U_1 (drives historical coupon usage **and** who the targeting model emails) and *brand affinity* U_2 (drives historical coupon usage **and** spend). Last year’s coupon-redemption count W is measured **before** the send and correlates with both treatment and outcome — by every heuristic it “looks like” a confounder. The data-generating model (exactly the gallery’s fourth panel):

$$\begin{gathered} U_1, U_2 \sim \mathcal{N}(0,1) \text{ (unobserved)}, \quad W = U_1 + U_2 + \nu, \quad \nu \sim \mathcal{N}(0,1), \\ T \sim \text{Bernoulli}(\sigma(0.8\,U_1)), \qquad Y = 20 + 6\,T + 3\,U_2 + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0,\,4^2). \end{gathered}$$

Trace the only indirect path: $T \leftarrow U_1 \rightarrow W \leftarrow U_2 \rightarrow Y$. It contains a **collider at** W (two arrowheads meet there), so it is **already blocked** — §3 formalizes this rule. There is *no open backdoor*: the naive estimate is unbiased before we touch anything. Conditioning on W *opens* the path. **Why does holding `coupon_history` fixed hurt?** $W \approx U_1 + U_2$, so *once you know* W, a high U_1 implies a low U_2 (they must sum to the observed W) — conditioning on the shared effect W makes the two hidden causes correlated. That freshly-manufactured U_1–U_2 link splices $T \leftarrow U_1$ onto $U_2 \rightarrow Y$, opening a T–Y path that was shut. This is ‘explaining away’ (§3) with one extra hop through the latents:

```
TRUE effect €6.00  *  corr(coupon_history, email) = +0.21, corr(coupon_history,
spend) = +0.41  -> 'looks like' a confounder
no controls                : €5.99   -> already unbiased (the collider at W
keeps the path shut)
controlling coupon_history : €5.22   -> conditioning OPENED the path - bias from
a 'safe' pre-treatment control
```

Read-out (M-bias). The naive estimate was already right; adding the pre-treatment, correlated-with-everything covariate moved it *away* from the €6 truth by the better part of a euro. This is the counterexample that kills the most common heuristic in applied work — **"control for everything measured before treatment."** A timestamp is not a criterion.

Two honest caveats before anyone panics and deletes covariates. First, magnitude: pure M-bias is usually **modest** next to confounder bias (the theoretical result that M-bias is typically small-magnitude — Ding & Miratrix 2015) — compare the sub-€1 shift here with the +€3.5 confounding bias coming in §4. Second, the common real-world tangle: when a variable is plausibly *both* a genuine confounder **and** an M-collider, adjusting is usually the lesser evil. The point is not "never control pre-treatment variables"; it is that **the decision comes from the graph, not the timestamp** — which is exactly why the rest of this notebook makes the graph explicit and lets *it* choose.

2 · Simulate a ground truth

`email` → `spend` with a **true effect of €6**. Around it: **`loyalty`** is a **confounder** (drives both email and spend → must control); **`responded`** is a **collider** (caused by both email and spend → tempting but poisonous); **`opened_email`** is a post-treatment descendant.

The data-generating model — exactly what `dgp.dag_control_demo` implements (defaults & seed in `src/cmp/dgp.py`). Loyalty $L \sim \mathcal{N}(0,1)$; an **unobserved** engagement trait $G \sim \mathcal{N}(0,1)$; $\sigma(z) = 1/(1+e^{-z})$:

$$
\begin{aligned}
T = \text{email} &\sim \text{Bernoulli}\big(\sigma(0.8\,L)\big) && \text{(assignment is confounded by loyalty)}\\
Y = \text{spend} &= 20 + 5\,L + 6\,T + 3\,G + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,\,4^2) && \text{(true ATE — average treatment effect: the 6)}\\
\text{opened\_email} &\sim \text{Bernoulli}\big(\sigma(1.5\,T + 0.7\,G)\big) && (T \to \text{opened} \leftarrow G \to Y)\\
\text{responded} &\sim \text{Bernoulli}\big(\sigma(1.2\,T + 0.05\,(Y - \bar{Y}))\big) && (T \to \text{responded} \leftarrow Y).
\end{aligned}
$$

L appears in both the treatment and the outcome equations — the backdoor that must be closed. Neither `opened_email` nor `responded` appears in the spend equation, so controlling for them can only manufacture bias: `responded` is the textbook collider (caused by T and Y directly), while `opened_email` is the *subtler* trap — its collider path runs through the **unobserved** G, which is why a DAG drawn without G cannot warn you (the falsification section below catches it from data).

```
TRUE effect of email on spend = €6.0

   email   loyalty  opened_email  responded      spend
0    1.0  1.101262           1.0        0.0  27.705363
1    0.0  0.338431           0.0        0.0  17.425501
2    0.0 -0.539972           0.0        0.0   9.602566
3    0.0 -1.260242           0.0        0.0   5.056443
4    1.0 -1.894621           0.0        0.0   7.175152
```

3 · Identify — let the DAG choose the adjustment set

So far "blocking a path" has been a metaphor. Before handing the graph to `pathmc`, we make it a **rule you can run by hand** — because the library below does nothing more than apply this rule exhaustively, and you should be able to audit it.

Rule 1 — when is a path blocked? (the d-separation rules). A *path* between T and Y is any sequence of edges connecting them, arrows pointing either way. Given a candidate control set Z, a path is **blocked** iff it contains at least one of:

$$\underbrace{A \to B \to C \;\text{ or }\; A \leftarrow B \to C \quad \text{with } B \in Z}_{\text{chain or fork: conditioning on the middle node closes it}}$$

or

$$\underbrace{A \to B \leftarrow C \quad \text{with } B \notin Z \text{ and no descendant of } B \text{ in } Z}_{\text{collider: closed by default — conditioning on } B \text{ (or its descendants) opens it}}$$

where a *descendant* of B is any variable reachable from B by following arrows forward. In words: conditioning on a **common cause** (fork) or an intermediate step (chain) removes the association that node transmits — compare customers at the *same* loyalty and the "loyal people get emailed and spend more anyway" signal disappears. Conditioning on a **common effect** (collider) does the reverse — it *creates* an association that was never there. The intuition is **explaining away**: among customers who responded, seeing high spend makes the email a *less necessary* explanation for the response, so email and spend become negatively related inside that group even if they are independent overall.

Rule 2 — the backdoor criterion. A set Z is an **admissible adjustment set** for the effect of T on Y iff

$$\text{(a) no element of } Z \text{ is a descendant of } T, \quad \text{and}$$
$$\text{(b) } Z \text{ blocks every path from } T \text{ to } Y \text{ that starts with an arrow into } T.$$

Paths starting with an arrow *into* T are the **backdoor paths** — the routes along which "who got treated" leaks information about the outcome. When (a) and (b) hold, the interventional quantity is recovered by the **adjustment formula**

$$P(Y \mid do(T{=}t)) \;=\; \sum_z P(Y \mid T{=}t,\, Z{=}z)\; P(z),$$

where $do(T{=}t)$ means "set T by intervention" (everyone gets the email) rather than "observe $T{=}t$" (the loyal self-select into it). Plain reading: *compare treated vs untreated within look-alike strata of Z, then average the strata back together with population weights.* Every regression adjustment in this notebook is a linear-model shortcut for that formula.

Run the algorithm by hand on our graph (the one declared to `pathmc` below). Enumerate every path from `email` to `spend`:

- `email` → `spend` — the causal path we are estimating. Leave it alone (blocking it, e.g. with a mediator, is §1c's trap);
- `email` ← `loyalty` → `spend` — starts with an arrow into `email` ⇒ a **backdoor**, open by default at the fork `loyalty` ⇒ any admissible Z must contain `loyalty`;
- `email` → `responded` ← `spend` — a **collider** at `responded` ⇒ already blocked; putting `responded` in Z would *open* it (and `responded` is a descendant of T, failing condition (a) twice over);

- `email` → `opened_email` — a dead end in the *declared* graph (no arrow onward to `spend`), so it lies on no T–Y path at all. Remember this one: it is exactly the blind spot the falsification section below exposes.

So $Z = \{\text{loyalty}\}$ is admissible (and minimal). That enumeration is **all** that `m.adjustment_sets` does — read its output below and check it against your own answer.

One more requirement the graph cannot show — positivity. The adjustment formula only averages strata where *both* arms exist: identification also needs $0 < P(T{=}1 \mid Z{=}z) < 1$ for every stratum z (**overlap**). Here it holds by construction — $\sigma(0.8\,L) \in (0,1)$ for every finite loyalty — but with a high-dimensional Z it fails silently; nb01's Step-4 propensity (the modelled $P(T{=}1 \mid Z)$) histogram (`est.propensity_scores` + `plots.overlap_plot`) is the standard check. Identification has two clauses — **no open backdoors *and* positivity** — not one.

Reading the `pathmc` spec below. Each line is *node ∼ its direct causes*, not a regression: `email ~ loyalty` declares the edge loyalty → email (loyalty is a cause of who gets emailed), not a regression of email on loyalty. So this block is exactly the four-node DAG we just enumerated by hand.

```
Admissible adjustment set(s) for email -> spend: [{'loyalty'}]
Identifiable: True

control for {'loyalty'}:  OK - no collider opened
control for {'loyalty', 'responded'}:  'responded' is a collider between 'email'
and 'spend'. Conditioning on it may open a spurious path and introduce bias.
control for {'loyalty', 'opened_email'}:  OK - no collider opened
control for {}:  OK - no collider opened
```

(Read the printout above as the machine doing the reasoning: it names `{loyalty}` as the one adjustment set that closes the backdoor, confirms the effect is identifiable, and — critically — *warns* when a candidate set includes `responded`, because that variable is a collider. This is the DAG logic from the intro, applied automatically.)

Step 0 · The classical read (no likelihood, no priors, no sampler)

Before any Bayesian machinery, ask what a competent analyst does here **without** it. In most of this cookbook that question opens a detour. Here it opens the *whole argument*, because the estimator this notebook needs is the plainest one there is: **ordinary least squares**, spend regressed on email plus an adjustment set Z,

$$\text{spend}_i \;=\; \beta_0 \;+\; \tau\,\text{email}_i \;+\; \gamma^\top Z_i \;+\; \varepsilon_i,$$

with $\hat{\tau}$, the coefficient on `email`, as **the estimand** — the €-per-email ATE §3 just wrote down, identified by the backdoor criterion under exactly the assumptions §3 listed (no open backdoor once Z is held fixed, plus positivity). No likelihood, no prior, no sampler; a point estimate and a confidence interval, in milliseconds.

Two fits, one estimator, and the only thing that changes is Z.

- $Z = \emptyset$ — the **naive** read. What you get by regressing spend on email and nothing else: the number a dashboard hands you.

- $Z = \{\text{loyalty}\}$ — the **backdoor-adjusted** read. The set `pathmc` endorsed above, and the one we enumerated by hand.

Both are OLS. Both are competently executed. One is right and one is wrong, and *the estimator has no idea which.*

The covariance choice, and why it is HC1 here. An interval on a wrong standard error is a confident lie, so the covariance assumption has to be argued, not defaulted. Our rows are **2,000 independent customers** measured **once**: there is no panel to cluster on (each customer contributes one row, so "within-cluster" correlation has nothing to correlate) and no time index to apply a HAC correction to. What remains is the cross-sectional worry — **heteroskedasticity**, i.e. residual variance that differs across customers (say, spend being more volatile among the loyal). `HC1` (the Huber–White sandwich with a small-sample correction) is robust to exactly that and is the honest cross-sectional default. Whether it *matters* on this data is an empirical question, not a rhetorical one, so the cell below prints a **Breusch–Pagan test** (its null is homoskedasticity) and the HC1-to-iid standard-error ratio, and we read the verdict off the print-out rather than assuming it.

```
THE ESTIMAND - the €-per-email ATE: the `email` coefficient of an OLS
fit on 2,000 customers. Same estimator, same data, same target.
Only the adjustment set Z changes.   [HC1 heteroskedasticity-robust]

  Z = {}        (naive)      € 9.45  [90% CI  8.94,  9.95]  SE 0.31
  Z = {loyalty} (adjusted)   € 6.09  [90% CI  5.69,  6.48]  SE 0.24

  Breusch-Pagan p = 0.94 (null: errors are homoskedastic)
  SE(HC1) / SE(iid) = 1.001

GRADE against the planted truth (€6 - a simulation-only luxury):
  Z = {}        (naive)      error +3.45   CI width €1.01   truth OUTSIDE
  Z = {loyalty} (adjusted)   error +0.09   CI width €0.79   truth INSIDE

What this 90% confidence interval does NOT say: that there is a 90% probability
the true effect lies inside it. It is a property of the *procedure* - 90% of
intervals built this way would cover the truth across repeated samples. This
interval either contains the truth or it does not. To get a probability *about
the effect itself* - the thing a go/no-go rule like P(lift > cost) >= 0.9
actually needs - you need a posterior. That is what the Bayesian section adds.
```

Read-out — the DAG did the work; the estimator merely executed it.

The adjusted fit lands on the planted €6 and its 90% interval covers it. The naive fit does not — and look at *how* it fails. It is not vague, hedged or evasive: its interval is of the same order of width as the adjusted one (the print-out gives both — roughly a euro wide), and it sits **several euros** away from the truth, which it excludes outright. That is the definition of **confidently wrong**: the omitted-variable bias from leaving `loyalty` out is a *fixed offset*, and a confidence interval quantifies **sampling noise around whatever the estimator converges to** — never the distance between that limit and the truth. Precision is not validity, and no standard error, however robust, will tell you which of these two numbers to believe.

Note also what HC1 bought: the Breusch–Pagan test does not reject homoskedasticity and the

HC1 standard error is within a fraction of a percent of the iid one — on *this* DGP the noise really is constant-variance Gaussian, so the robust sandwich changed nothing. We keep it anyway, and say so plainly: you cannot know that in advance without checking, HC1 costs nothing when errors are homoskedastic, and it saves you when they are not. Insurance that turns out unnecessary was still worth buying.

Now the sentence this whole notebook exists to earn. The gap between those two lines was produced by a choice made *before* any estimator ran — which variables to hold fixed — and it is the entire difference between a defensible number and a €3-per-customer fantasy. **No amount of Bayesian machinery closes that gap.** A prior cannot tell you that `loyalty` is a confounder; a posterior cannot detect that a backdoor is open; MCMC will converge beautifully on the wrong estimand. *The DAG does the causal work; the estimator only executes it.* §4–5 now runs the same OLS across five adjustment sets to price that claim, and **5x** holds the classical answer against the Bayesian posterior on the same set, so you can see exactly what the machinery adds — and what it cannot.

Finally, the boundary printed above, which is the same in every notebook of this cookbook: a 90% confidence interval is a property of the *procedure*, not a probability about the effect. The go/no-go rule §6 actually uses — $P(\text{lift} > \text{cost}) \geq 0.9$ — is a statement *about the effect itself*, and that sentence has no frequentist translation. It needs a posterior. That, and not a better point estimate, is what §4's Bayesian layer is for.

4–5 · Estimate & Validate — every control choice, in €

Now we make the abstract warnings concrete and expensive. We estimate the same email→spend effect **five ways**, each time controlling for a different set of variables, and compare to the known true €6:

- **nothing** (naive) — leaves the loyalty confounding in, so it's biased *up*;
- **{loyalty}** — the DAG-endorsed set; should land on €6;
- **{loyalty, responded}** — adds a *collider*, which opens a spurious path;
- **{loyalty, opened}** — adds a *post-treatment descendant*, another no-no;
- **everything** — the "throw it all in" instinct, which stacks the errors.

The punchline is that **the sign and rough size of each bias is predictable from the graph *plus the signs of its edges before* you run anything** (here loyalty raises both emailing and spend): omitting a confounder biases up; conditioning on a collider or a post-treatment variable biases in the direction of the association it induces. Controlling for more is *not* safer.

The fitted model, in symbols — the same OLS regression five times, differing only in the control set W:

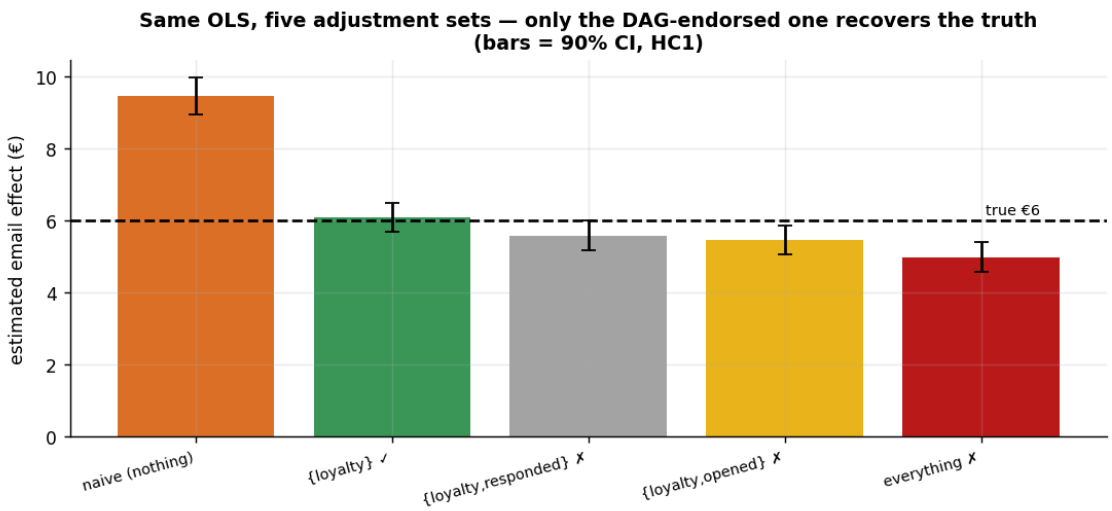
$$\text{spend}_i = \beta_0 + \hat{\tau}\,\text{email}_i + \gamma^\top W_i + \varepsilon_i,$$

reading off $\hat{\tau}$ for each choice of $W \in \{\emptyset,\ \{L\},\ \{L, \text{responded}\},\ \{L, \text{opened}\},\ \{L, \text{responded}, \text{opened}\}\}$. Same estimator, same data — *only the adjustment set changes*, so every difference between the bars below is pure identification, not modeling. This is **Step 0's fit, run five times**: the same `cl.ols` call, the same `email` coefficient, the same HC1 covariance (argued above: independent customers, one row each, possible heteroskedasticity), so every bar carries a 90% confidence interval and no bar is a bare point estimate. Watch what the intervals do *not* do: they are all of similar width, and none of them flags the bar it belongs to as biased.

```
OLS `email` coefficient * HC1 heteroskedasticity-robust * n = 2,000

  adjustment set                  €    90% CI           error   width   covers €6?
  naive (nothing)              9.45   [ 8.94,  9.95]  +3.45   1.01   NO
  {loyalty} [ok]                  6.09   [ 5.69,  6.48]  +0.09   0.79   yes
  {loyalty,responded} [x]       5.58   [ 5.15,  6.00]  -0.42   0.84   NO
  {loyalty,opened} [x]          5.45   [ 5.04,  5.85]  -0.55   0.81   NO
  everything [x]                4.97   [ 4.55,  5.40]  -1.03   0.84   NO

1/5 of the five 90% intervals cover the truth, and the widths span only
€0.79-€1.01:
the wrong sets are not vaguer than the right one, they are just as precise -
about a different number.
```



How to read this. Only the green bar — `{loyalty}`, the set the DAG endorsed — lands on the true €6, and it is the only one whose interval covers it (the `covers €6?` column). The naive bar overshoots (confounding left in); adding the collider `responded` or the hidden collider `opened_email` (see the falsification test below) *pulls the estimate away* even though we already had the right confounder; and "everything" is the worst of all.

Now read the **widths**, because that is where the lesson bites. Every interval is roughly the same size — the wrong sets are not *hedging*, they are not *noisy*, they are not *flagged*. They are precise, and they are precise about the wrong number. Four of these five fits are a confident lie, and nothing inside the regression output — not the standard error, not the R^2, not the robust covariance we so carefully argued for in Step 0 — distinguishes them from the fifth. The moral for a practitioner is blunt: **the model can't tell you it's biased — the bars all look like plausible numbers, with plausible error bars.** The only defence is choosing the control set from the causal graph *before* looking at the estimates, which is precisely what makes the answer defensible to a skeptic.

Depth check — predict the bars before running them

§4–5's preamble claimed the sign *and rough size* of each bias is readable off the graph. That deserves a derivation, not an assertion. Both wrong directions follow from two pocket formulas.

1 · Omitted confounder ⇒ omitted-variable bias (OVB). Regress spend on email alone under the §2 DGP ($Y = 20 + 5L + 6T + 3G + \varepsilon$, with the engagement trait G independent of T). The naive coefficient converges to

$$\hat{\tau}_{\text{naive}} \xrightarrow{p} \tau \ + \ \beta_L \left(\mathbb{E}[L \mid T{=}1] - \mathbb{E}[L \mid T{=}0]\right), \qquad \beta_L = 5,$$

the classic OVB formula specialized to a binary treatment: **bias = (effect of the omitted variable on the outcome) × (its imbalance across arms)**. Both factors are positive here — loyalty raises spend ($\beta_L = 5 > 0$) and loyal customers get emailed more (positive imbalance) — so the graph plus edge signs says *biased up*, and the DGP coefficients say *by how much.* The cell below computes the prediction and holds it against the naive bar.

2 · Conditioned collider ⇒ explaining-away, with a sign rule. For a collider receiving $C \leftarrow a\,T$ and $C \leftarrow b\,Y$, conditioning on C induces a non-causal T–Y association of sign $-\,\mathrm{sign}(a\,b)$ — the minus sign is explaining-away made quantitative: if T and Y both *push* C *up* ($a, b > 0$), then within a fixed-C stratum a high T must be offset by a low Y, so they read as negatively related. For `responded`, $a = 1.2 > 0$ and $b = 0.05 > 0$, so the induced association is **negative** — the pocket rule: *both arrows into the collider positive ⇒ conditioning biases the estimate down.* That is the direction of all three × bars. (The M-bias example of §1c runs on the same explaining-away algebra, just with one extra hop through the latents.)

```
loyalty imbalance across arms      DeltaL = +0.691
predicted naive bias    beta_L*DeltaL = 5 x 0.691 = +3.46
observed naive bias     (naive bar - truth)   = +3.45   -> they agree to €0.01

collider sign rule: a = +1.2, b = +0.05  =>  sign(-a*b) < 0 - and indeed all
three [x] bars err downward: -0.42, -0.55, -1.03
```

The graph gave the *directions* before any data existed; the DGP's coefficients turned direction into magnitude; and the simulation confirms the OVB prediction to within a few cents. That is the standard to hold an observational read to: **if you cannot pre-state the sign of the bias a proposed control would create or remove, you are not ready to defend the estimate** — and when your predicted sign and the data disagree, suspect the graph (the falsification test below makes that suspicion operational).

When the machine is wrong — falsify the DAG against the data

Look back at Step 3: pathmc **green-lit `{opened_email, loyalty}`** ("OK — no collider opened"), yet the bar above shows that exact set is **biased** (−€0.55). That is not a pathmc bug — it is the single most important caveat about DAG-based methods: **the machine reasons only about the graph *we drew.*** Our graph declares `opened_email ~ email`, but the true process has a **hidden engagement trait** driving both `opened_email` and `spend` — making `opened_email` a collider pathmc cannot see, so conditioning on it opens `email` → `opened_email` ← `trait` → `spend`.

You can catch this *without* knowing the hidden trait: every DAG makes **testable conditional-independence claims**. Our declared graph implies `opened_email` $\perp$ `spend | {email, loyalty}` — so test it against the data:

```
Declared-DAG conditional-independence claims tested: 4  *  violations: 1
           x     y conditioning_set      p_value
opened_email spend   email, loyalty 1.314202e-19
```

The data **rejects** `opened_email` $\perp$ `spend | {email, loyalty}` (p $\approx$ 1e-19): a testable implication of the graph we drew is false, so the graph is misspecified. *That* is why `{opened_email, loyalty}` comes out biased even though pathmc approved it — the tool was faithful to a wrong graph. **The lesson: a causal DAG is an assumption. Let the machine pick the adjustment set from the graph, but *falsify the graph against the data* (here, `test_implications`) before you trust its verdict.**

4b · The same trap without a regression — analyzing only responders

Nobody on a marketing team types `responded` into a regression and feels clever about it. What teams do instead — constantly — is **filter**: *"let's measure the email's effect among customers who engaged."* The two are the same act: **subsetting a dataset IS conditioning on the subsetting variable.** Keeping only `responded == 1` conditions every downstream estimate on the collider $T \rightarrow \text{responded} \leftarrow Y$, opening the same non-causal path as §4's grey bar — only now it is invisible, because no model summary will ever list `responded` as a covariate.

```
filtered to responders: 1,194 of 2,000 customers (60%) - hardly a niche slice
true effect €6.0   *   all three fits: OLS, HC1, 90% CI

  full sample,      {loyalty}: € 6.09  [ 5.69,  6.48]  SE 0.24  -> covers €6
  responders only, nothing  : € 8.73  [ 8.04,  9.42]  SE 0.42  -> MISSES €6
  responders only, {loyalty}: € 5.71  [ 5.18,  6.24]  SE 0.32  -> covers €6

  ^ the RIGHT confounder, correctly HC1-ed, honest interval - and its point
    estimate is -0.29 off. In THIS sample the interval still covers €6:
    nothing looks wrong. One sample cannot tell a modest bias from noise -
    only repeated sampling can. Re-run it on fresh data:

responders-only + {loyalty}, 30 fresh samples: mean bias -0.48 (sd 0.40),
i.e. 6.5 standard errors from zero - a persistent shortfall, not sampling
luck: no covariate adjustment inside the subset can close a path that the
filter itself opened.
```

Read-out. Inside the responders subset the naive read is badly off, and — the important line — handing the filtered analysis the *exactly correct* confounder still leaves a bias that does not average away across fresh samples. Note carefully how it hides: in the single sample above, the responders-only estimate is only a few tens of cents low and its 90% interval **still covers the truth**, so the fit looks blameless. It is the fresh-sample sweep that convicts it — the shortfall reappears sample after sample, many standard errors from zero, exactly the signature (§5c) that separates a *bias* from *noise*. A confidence interval on a selected sample is honest about sampling error and silent about the selection. The problem is not an omitted variable; it is an **opened path**, and the filter itself opened it. Recognize the pattern in the wild, because it rarely announces itself:

- **converter-only attribution** — “of the customers who purchased, which touchpoints did they see?”;
- **opened-email lift reads** — “the subject line’s effect among opens”;
- **survey-based incrementality** — only responders answer the survey;
- **“active users” dashboards** — activity is caused by both the campaign and the outcome it proxies.

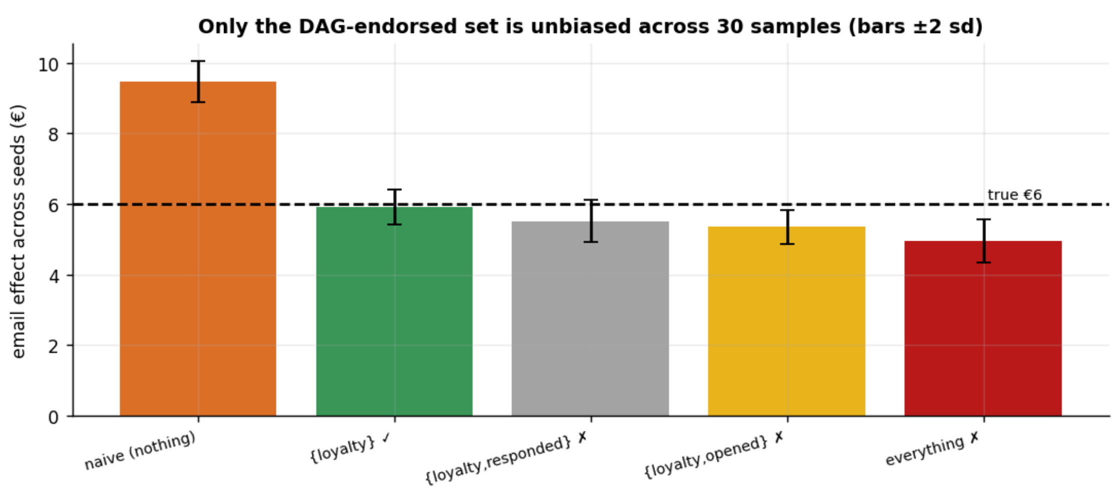
The fix is never a better covariate *inside* the subset — it is refusing the filter (estimate on everyone the campaign could have touched) or modeling the selection explicitly.

5b · Recovery across many seeds — unbiased *and* calibrated?

A single sample can land a little off; the real test of an estimator is whether it is **centred on the truth over repeated samples** (unbiased, not just lucky here) and whether its intervals **cover** the truth at their stated rate. We refit on many fresh samples and check both.

```
naive (nothing)           mean € 9.47  bias +3.47  sd 0.30
{loyalty} [ok]               mean € 5.92  bias -0.08  sd 0.25
{loyalty,responded} [x]     mean € 5.51  bias -0.49  sd 0.31
{loyalty,opened} [x]        mean € 5.35  bias -0.65  sd 0.25
everything [x]              mean € 4.96  bias -1.04  sd 0.31

The DAG-endorsed set is centred on the truth (bias -0.08 across 30 seeds -
sampling noise, not a fixed offset); the collider / over-control sets carry a
PERSISTENT bias
(a fixed offset that does NOT shrink across seeds, unlike sampling noise) -
exactly what a single sample can hide.
```



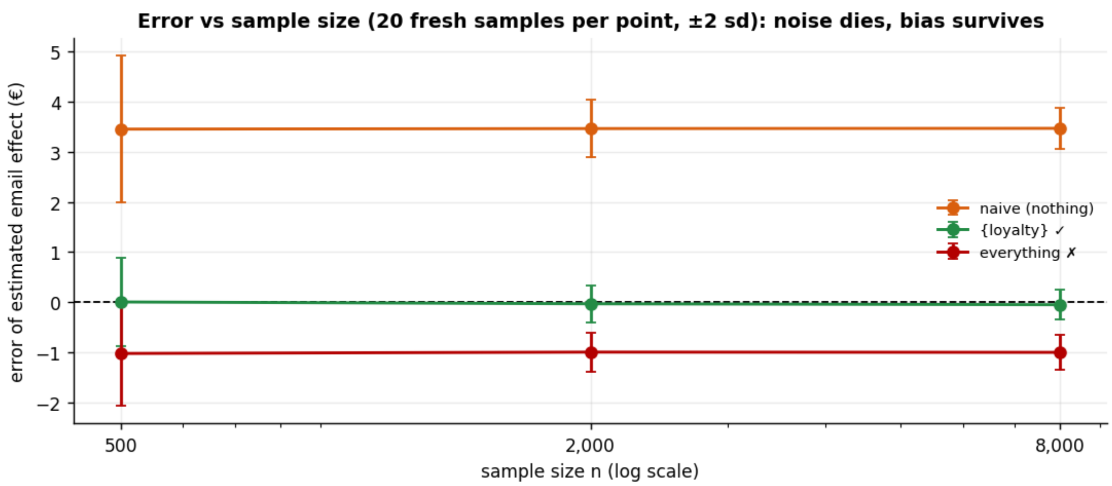
5c · “But we have millions of rows” — more data does not fix a bad control set

The bars above separate noise from bias at a single sample size; the print-out *claims* the wrong sets’ offsets would not shrink with more data. Make the claim formal: for control set W the OLS estimate obeys

$$\hat{\tau}_W \xrightarrow{p} \tau + b_W,$$

where b_W is a **fixed asymptotic bias** determined only by identification — which backdoor paths W leaves open (the naive set) or which collider paths it opens (the kitchen-sink set). As $n \to \infty$, sampling noise vanishes but b_W does not move an inch: **more data buys precision, never validity.** We verify by sweeping the sample size for three control sets, many fresh samples per point:

```
naive (nothing)    error: n=500 -> +3.45+/-0.73   n=8,000 -> +3.47+/-0.21
{loyalty} [ok]        error: n=500 -> +0.01+/-0.45   n=8,000 -> -0.05+/-0.15
everything [x]       error: n=500 -> -1.02+/-0.52   n=8,000 -> -1.00+/-0.17
```



Read-out. Every error bar narrows as n grows — but only the green line narrows **around zero**. The naive and kitchen-sink lines narrow around the wrong number: at $n = 8{,}000$ they are *more confidently wrong* than at 500. This is the rebuttal to the pitch every CMO eventually hears — "our platform observes millions of customers, so the numbers are rock-solid." Volume shrinks the error bars; if the adjustment set is wrong, it shrinks them around the artifact. The first question for any vendor deck is not *how many rows* but ***what did you control for — and what did that choice condition on?***

5x · Point estimate vs posterior — what the Bayesian layer buys, and what it cannot

Everything above was classical: one `cl.ols` call per adjustment set, five point estimates, five confidence intervals. §4's Bayesian model — the `pathmc` fit that §6's sensitivity grid runs on — targets the **same estimand** (`email`'s coefficient in the spend equation) under the **same adjustment set** ($Z = \{\text{loyalty}\}$, because the declared spec says `spend ~ email + loyalty`) with **weak priors** and the same 2,000 rows. So the two should agree, and if they do we should say so without flinching.

But agreement on the endorsed set is only half the account. This notebook's thesis is that identification, not inference, is where the causal content lives — and §5's read-out *asserted* that a

Bayesian refit on the kitchen-sink controls "would hand you an interval that looks essentially as tight, centred on the biased number." An assertion is not a demonstration. So the cell below fits **the same Bayesian machinery a second time on the wrong adjustment set** — spend on email plus loyalty *plus the collider and the descendant* — with the same priors and the same sampler, and reports its posterior and its decision probability alongside the others. Three columns, one estimand, and the only difference that matters is Z.

```
  arm                         adjustment set Z          € 94% interval  width
  classical OLS (94% CI)     {loyalty} - endorsed   6.09 [ 5.63, 6.54]  0.91
  Bayes posterior (94% HDI) {loyalty} - endorsed   6.09 [ 5.66, 6.57]  0.90
  Bayes posterior (94% HDI) everything - WRONG     4.97 [ 4.49, 5.44]  0.96

Z = {loyalty}: the two arms differ by €0.006 on the point estimate
  (0.03 classical SEs), and their intervals differ in width by €0.005.
  The truth (€6) is inside BOTH.
Z = everything: the posterior sits €-1.03 from the truth, which
  falls OUTSIDE its 94% HDI - and it is just as tight (€0.96 vs €0.90 wide).
  Nothing in the output announces the bias.

THE DECISION QUANTITY - what no confidence interval can produce: P(tau > €5)
  Bayes, Z = {loyalty}  (right) : P(tau>0)=1.000 * P(tau>cost)=1.000 -> GO
  Bayes, Z = everything (WRONG) : P(tau>0)=1.000 * P(tau>cost)=0.448 -> NO-GO
  classical, either set         : does not exist - a CI is not a
                                  probability about tau.

kitchen-sink convergence: max r-hat 1.000 - min ESS 4047 - divergences 0
```

The honest verdict — three beats, in the order they matter.

1 · On the endorsed set, they agree. Say it plainly. The classical OLS estimate and the Bayesian posterior mean differ by a rounding error — a fraction of one standard error, as the print-out quantifies — and the 94% credible interval is, to within a couple of cents, the 94% confidence interval. That is not a disappointment; it is the point. With 2,000 rows and priors this weak, the posterior *is* the likelihood, and the likelihood *is* the least-squares fit. **The Bayesian layer did not move the number, because the number was never where the difficulty lay.** Anyone who arrived hoping that "going Bayesian" would improve the estimate has just been shown, on their own data, that it does not.

2 · What the posterior does buy — precisely one thing here, and it is the thing the business asked for. Look at the last block: $P(\tau > \text{cost})$. §6 and §7 run the campaign decision on $P(\text{lift} > \text{cost}) \geq 0.9$, and that is a probability **about the effect**, natively a posterior quantity. The classical arm cannot produce it — not with a tighter CI, not with a bootstrap (which approximates the *sampling distribution of the estimator*, not a distribution of belief over τ), not with a p-value. There is no number to compare against the 0.9 bar; the row is empty by construction. That single column is the whole earned payoff of the Bayesian machinery in this notebook, and it is a real one: it turns "the point estimate beats €5" into a stated risk of being wrong.

Be equally precise about what does *not* belong in that column. §6's $\gamma{\cdot}\delta$ **sensitivity contour** and the **E-value** are arithmetic on the *observed ATE* — `adjusted_ATE = observed_ATE` $-\ \gamma{\cdot}\delta$, and RR $\approx \exp(0.91d)$ — and would run identically off the classical point estimate. They are sensitivity analysis, not Bayesian inference. Crediting them to the posterior would be padding the bill.

3 · What Bayes did NOT buy — and this is the notebook's thesis, now demonstrated rather than asserted. The third row of the table is the same Bayesian machinery — a PyMC linear model with priors as weak as `pathmc`'s, sampled by the same NUTS, on the same 2,000 rows — pointed at the kitchen-sink adjustment set. Its posterior is **tight, well-mixed, convergent (r-hat and ESS are printed, and they are immaculate), and centred on the wrong number**; the truth falls outside its 94% HDI; its credible interval is essentially as tight as the correct model's (€0.96 vs €0.90 — the extra six cents buys you nothing); and it emits a perfectly well-formed decision probability $P(\tau > \text{cost})$ — which, being computed about a biased estimand, points at the *opposite decision* from the correct model. Read the two GO/NO-GO verdicts side by side: same machinery, same data, same rule, opposite call. Nothing in the diagnostics complains, because **nothing is wrong with the inference** — the inference is faithfully reporting the posterior of a quantity that is not the causal effect.

A collider is a collider under any inference paradigm. Priors do not close backdoor paths; posteriors do not detect open ones; MCMC converges just as happily on a biased estimand as on an unbiased one, and reports its uncertainty just as confidently. **Bayes buys you a decision quantity; it does not buy you identification.** Identification came from the DAG in §3, and if the DAG is wrong — as §5's falsification test showed it can be — then the posterior inherits the error and dresses it in a credible interval. That is why this notebook, alone in the cookbook, spends its length on a graph instead of a model.

6 · Decide, in euros — the stakes of the control choice, with a robustness statement

Two parts. **(a) The euro stakes.** Getting the control set wrong isn't academic: budgeting a campaign on the naive (confounded) estimate books *phantom* incremental revenue, which we price out below across a realistic campaign — the concrete cost of skipping the DAG. **(b) The robustness statement.** Identification is untestable, so we bound the residual risk: how strong would an *unobserved* confounder have to be to overturn the conclusion? `pathmc.sensitivity` and the E-value are **two independent parameterizations (two different ways of dialing the same knob) of the same worry**, not one summarising the other. The first is an **additive $\gamma{\cdot}\delta$ contour in raw € units**: it sweeps a hidden confounder's treatment- and outcome-associations (γ, δ) and maps where the adjusted effect `observed_ATE` $-$ $\gamma{\cdot}\delta$ tips through zero — i.e. where $\gamma{\cdot}\delta \approx$ **the observed ATE** (the grid is widened so that zero-crossing is actually reachable). The second is VanderWeele's **E-value**, a single number on the **risk-ratio scale**: for a continuous euro outcome the effect is first standardized (rescaled into standard-deviation units — the code below prints the d it uses) and mapped to an approximate risk ratio via RR $\approx$ exp(0.91·d), and the E-value reports how strong the confounder's risk-ratio associations would have to be. Because the two live on **different scales** (raw € vs risk ratio), the E-value does *not* summarise the $\gamma{\cdot}\delta$ contour, and the euro effect must *not* be plotted on the E-value's axis.

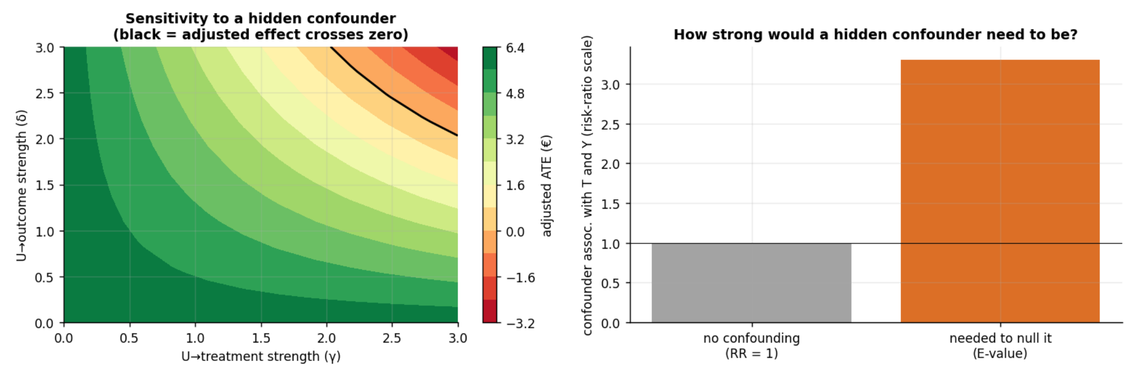
And because the fit is Bayesian, a third part follows the code: we take the posterior **5x** already sampled and restate the GO/NO-GO as a probability instead of a point comparison. (Both the fit and the €5 break-even `COST` were set in 5x; §6 spends them.)

```
Observed adjusted ATE €6.09 (true €6.0)  *  standardized d = 0.73 sd (spend sd
€8.33)  *  E-value ~ 3.30
An unmeasured confounder would need ~3.3x association (risk-ratio scale) with
```

```
BOTH email and
spend to explain the effect away - ask the domain expert whether anything that
strong is plausible.

Euro stakes: budgeting on the naive €9.45 instead of the DAG-endorsed €6.09
overstates incremental
revenue by €168,031 across a 50,000-customer campaign - phantom lift that would
justify overspending on the email.

Break-even €5.00/email * E-value vs break-even ~ 1.50 (how strong a hidden
confounder must be to drag the DAG-endorsed effect below cost).
  naive (nothing)           € 9.45  ->  GO   (lift > cost)
  {loyalty} [ok]              € 6.09  ->  GO   (lift > cost)
  {loyalty,responded} [x]     € 5.58  ->  GO   (lift > cost)
  {loyalty,opened} [x]        € 5.45  ->  GO   (lift > cost)
  everything [x]              € 4.97  ->  NO-GO (lift <= cost)
Same data, 1/5 control sets fall below the €5.00 break-even: over-controlling
drags the true €6.0
effect below cost, so the kitchen-sink set would wrongly KILL a campaign the
DAG-endorsed {loyalty} set green-lights - the control choice is a business call,
not a technicality.
control model convergence: max r-hat 1.000 - min ESS 5174 - divergences 0
```

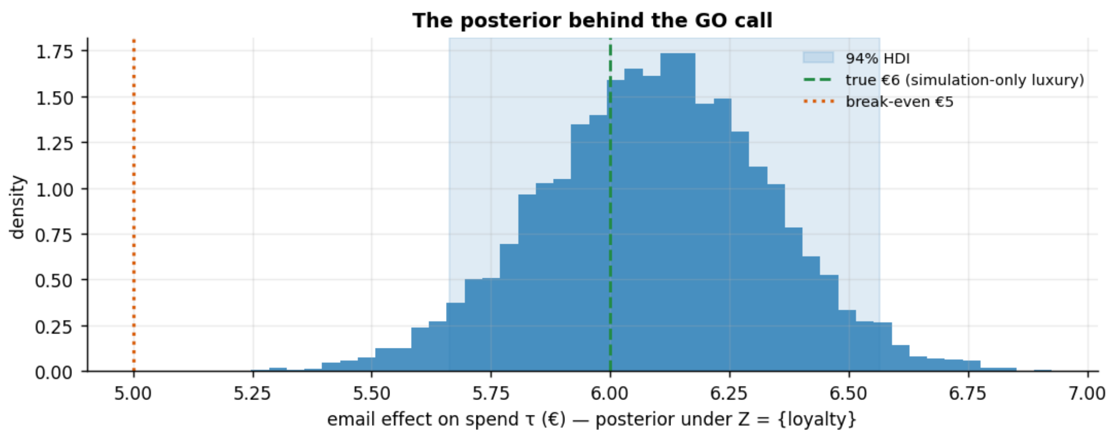


The Bayesian read — how sure are we, and of what?

Every GO/NO-GO in the table above compared a *point estimate* to the €5 break-even — thin ice for a Bayesian cookbook. The `m.fit(...)` in 5x, which also powered the sensitivity grid, is a full posterior over every structural coefficient, so we already own the entire distribution of the email effect **under the endorsed set** $\{loyalty\}$. The honest decision statement replaces "point estimate > €5" with the two probabilities 5x printed: $P(\tau > 0)$ — is there any effect — and $P(\tau > \text{cost})$ — does it pay. Here we plot the distribution behind them.

On the convergence line printed above (`control model convergence: max r-hat … · min ESS … · divergences …`): **R-hat** near 1.00 means the independent sampler chains agree ($\leq$ **1.01 passes**); **ESS** (effective sample size) is the number of *independent* posterior draws behind each estimate (a few hundred is plenty here); **0 divergences** means the sampler hit no numerical pathologies. Under the FAST profile the short chains can nudge R-hat to ~1.02 and trigger a benign "problems during sampling" notice that clears in a FULL run.

```
posterior mean €6.09   *   94% HDI [€5.66, €6.57]   *   classical OLS (HC1)
€6.09 [€5.69, €6.48]
P(tau > 0) = 1.000   *   P(tau > €5 break-even) = 1.000
Decision, restated honestly: GO with P(lift > cost) = 1.00 under the DAG-
endorsed adjustment set - a probability, not a point comparison.
```



Read-out. Identification (Steps 3–5) chose *which* number the posterior concentrates on; the posterior says *how sure* to be of that number. Here the whole 94% HDI sits above the €5 break-even, so $P(\text{lift} > \text{cost})$ prints at essentially 1 and the GO is not riding a knife's edge — if the HDI straddled the cost line, the same plot would say "test before you spend" (nb01's straddler logic). One warning label, and it matters: **this width is honest only under the endorsed adjustment set.** 5x refitted this same machinery on the kitchen-sink controls and it handed back an interval essentially as tight (the printed widths differ by six cents) — centred on the biased number, with a decision probability that flips the call. A tight posterior around a collider-biased point is *confidently wrong*: precision quantifies sampling noise, never identification error. That is §4's "the model can't tell you it's biased," now in Bayesian dress.

6b · The stakes on real data (LaLonde) — good vs bad adjustment

This isn't an academic worry, and it isn't confined to toy graphs. The famous **LaLonde (1986) / Dehejia–Wahba** job-training data pairs a randomized experiment — benchmark effect on 1978 earnings ≈ $1,800 — with an observational comparison group drawn from population surveys, so observational adjustment choices can be *scored* against an experimental answer. The cell below prints three reads of the same observational data:

- **naive** (no controls) — the trainees were far poorer than the survey comparison group before the program existed, so the raw gap is dominated by selection;
- **bad adjustment** — demographics only (age, education, race, marital status, degree), *dropping* the two pre-program earnings years `re74`/`re75`. It looks responsible — six pre-treatment

covariates! — but Dehejia & Wahba's core finding is that the pre-earnings variables are the **load-bearing confounders**;
- **good adjustment** — the same regression *plus* `re74`/`re75`, the full standard pre-treatment set.

The data is fetched from a public URL (you provide nothing) and the section is gated on `CMP_REAL=1` so offline runs skip it; the committed notebook ships with the gated branch **executed**, and the read-out below refers to those printed numbers.

```
Effect of job training on 1978 earnings (observational comparison group, n=614):
  naive - no controls                          : $   -635
  BAD adjust - demographics, NO re74/re75      : $  1,164
  GOOD adjust - demographics + re74/re75       : $  1,548
  experimental benchmark (RCT reference)       : ~ $1,800
why the naive fails: 1974 earnings were $2,096 (future trainees) vs $5,619
(comparison group) - the groups were never comparable.
```

Read-out (gated run). The naive comparison does not merely miss the benchmark — on this composite it comes out **negative**, "training *reduced* earnings," because the comparison group out-earned the trainees years before the program existed (the last printed line quantifies the pre-program gap). Adjusting for demographics alone repairs the sign but still lands well short of the good specification; restoring the two pre-program earnings years moves the estimate within a few hundred dollars of the ≈ $1,800 experimental answer. Same data, three defensible-*sounding* specifications, three different stories — and choosing among them is exactly the §3 skill: prior earnings are the strongest plausible common cause of *enrolling in the program* and *earning later*, so the backdoor criterion demands them in Z. Dropping a load-bearing confounder is the mirror image of adding a collider: both are identification failures no estimator or sample size repairs (§5c). *(If you re-executed this notebook without `CMP_REAL=1`, the numbers referenced here are in the committed output; re-run the gated branch before quoting them elsewhere.)*

7 · From the toy graph to your CRM — a practitioner's covariate checklist

The intro promised to tell you which columns of *your* campaign table belong in the model. The toy variables map one-to-one onto the fields every marketing dataset carries:

variable (as measured)	likely causal role	action
loyalty tier, RFM score computed **before** send, region, seasonality, prior-quarter spend	confounder — drives both targeting and spend	**control**
opens, clicks, site visits **after** the send	mediator — or hidden collider via latent engagement (§4's `opened_email`)	**exclude** when the question is the total effect
responded / converted / unsubscribed	collider (post-outcome)	**exclude — and never filter on it** (§4b)
survey participation, post-campaign app install	selection collider	**exclude**; distrust any panel filtered on them

variable (as measured)	likely causal role	action
pre-treatment *proxy* of an unmeasured trait (e.g. historical open-rate, coupon history)	it depends — draw the graph	M-bias risk if it is a common effect of latents (§1c)

Pocket rule: **pre-treatment AND a plausible common cause of T and Y $\Rightarrow$ in; caused by T $\Rightarrow$ out; anything else $\Rightarrow$ draw the DAG.** A timestamp is a heuristic; the graph is the criterion — and when in doubt between "possible confounder" and "possible M-collider," §1c's magnitude argument says adjusting is usually the lesser evil.

The one-paragraph decision

> We estimate the email's effect on spend adjusting for **`{loyalty}` only** — the set the backdoor criterion selects and the only candidate that survives falsification against the data (`opened_email` fails its implied-independence test, so it stays out even though the *declared* graph green-lit it). Under that set the effect is $\approx$ €6 with the 94% HDI printed above, clearing the €5 break-even with $P(\text{lift} > \text{cost}) \approx 1$: **GO**. Robustness: a hidden confounder would need roughly a 3$\times$ risk-ratio association with *both* treatment and outcome to null the effect, and about 1.5$\times$ to drag it below break-even — the second number is the one to debate with the domain expert. The control choice is not a technicality: budgeting on the naive read books $\approx$ €170k of phantom revenue on a 50,000-customer campaign, and the kitchen-sink set flips this true-GO campaign to NO-GO. Residual risk is the graph itself, so the follow-up is not a bigger model — it is re-running `test_implications` on each quarter's fresh data and re-drawing the DAG the moment the data rejects an implication. Note what does *not* appear in this memo as a safeguard: the estimator. Plain OLS on the endorsed set (Step 0) and the Bayesian posterior (5x) return the same number to the cent; the posterior earns its place only by producing $P(\text{lift} > \text{cost})$, and 5x showed that the very same posterior, run on the kitchen-sink controls, would have reported that probability just as confidently for the wrong estimand.


```
{
  "adjustment_set": [
    "loyalty"
  ],
  "ate_hat_classical_ols_hc1": [
    6.09,
    [
      5.69,
      6.48
    ]
  ],
  "ate_hat_posterior_mean": 6.09,
  "hdi94": [
    5.66,
    6.57
```

```
  ],
  "p_gt_zero": 1.0,
  "p_gt_cost": 1.0,
  "e_value_null": 3.3,
  "e_value_breakeven": 1.5,
  "phantom_revenue_naive_eur": 168031,
  "decision_flipping_sets": [
    "everything [x]"
  ],
  "p_gt_cost_if_kitchen_sink_controls": 0.448,
  "dag_falsification": {
    "n_tests": 4,
    "n_violations": 1
  }
}
```

8 · Caveats

- **The graph is an assumption.** `pathmc` gives the correct set *for the DAG you drew.* A missing or reversed edge changes the advice — the DAG must encode defensible domain knowledge.
- **Colliders hide in "obvious" controls.** "engaged", "responded", "opened", "clicked" are almost always post-treatment; excluding them feels wrong to practitioners and is exactly right.
- **Front-door as a fallback.** If a T–Y confounder is genuinely unmeasured but a fully-mediating measured variable exists, the front-door adjustment can still identify the effect — a tool to keep in reserve (notebook 04 builds the mediation machinery front-door leans on; full front-door identification needs assumptions beyond nb04's scope).
- **Sensitivity ≠ proof.** A high E-value means "robust to plausible hidden confounding," not "unconfounded." Pair it with the expert's judgement about what could be missing.